Fiscal Year:	FY 2022	**Task Last Updated:**	FY 11/05/2025
PI Name:	Pinsky, Lawrence Ph.D.		
Project Title:	Timepix-based Radiation Environment Monitor for SpaceX (TREMS)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	pinsky@Central.UH.edu	**Fax:**	FY 713-743-3589
PI Organization Type:	UNIVERSITY	**Phone:**	713-743-3552
Organization Name:	University of Houston		
PI Address 1:	Physics Department; 3507 Cullen Blvd		
PI Address 2:	S & R 1; Rm 617		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77204-5005	**Congressional District:**	18
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	TRISH--EXPAND
Start Date:	05/01/2021	**End Date:**	09/30/2021
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	SpaceX		
Flight Assignment:	SpaceX Inspiration4 NOTE: End date changed to 9/30/2021 per E. Urquieta/TRISH; original end date was 4/30/2022 (Ed., 10/20/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Chancellor, Jeffery Ph.D. (Louisiana State University)		
Grant/Contract No.:	NNX16AO69A-EXP0004		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	EXPAND Project The aim of this cooperative project between the University of Houston (UH) and Louisiana State University (LSU) is to characterize the ionizing radiation environment applicable to humans on orbit using an active hybrid radiation-imaging pixel semiconductor detector employing the Timepix technology from the CERN (European Organization for Nuclear Research)-based Medipix2 Collaboration. This project is a follow-on to a previously Translational Research Institute for Space Health (TRISH)-funded successful UH project to measure the radiation environment onboard regular commercial flights in polar regions. This was done with an eye towards using that environment as an analog to the environment onboard the International Space Station (ISS). While the demonstration succeeded in showing that the dose equivalent rates and Linear Energy Transfer (LET) spectra were comparable, and the hardware functioned as designed, the interest in using that environment for further research did not materialize.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	During the work on this project, the Louisiana State University (LSU) team designed a small aluminum case using 3-D printing technology for the MiniPIX-TPX3 electronics, plus a Timepix3 circuit board that can be used in future projects. During the development process, the team was able to deliver a potential flight-unit. But this potential fight-unit required fulltime use of one of the USB ports, which was rejected by SpaceX. Therefore, the hardware could not be flight-certified.
Bibliography Type:	Description: (Last Updated:)